

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

26th November 2019

Subject SA2 – Life Insurance

Time allowed: 3 Hours 15 Minutes (14.45 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** You work at an established, mid-sized Indian life insurer. The historical profitability track record of the Company has been moderate.

Over the last few years, the Company has witnessed strong new business growth, mainly attributable to strong sales through unit-linked products. The Company anticipates this trend to continue in the near future. As unit-linked products are capital intensive, the Company's business plan projects a potential capital injection in about three years' time.

- i)** Outline why the unit-linked business tends to be capital intensive in the Indian context. (5)

The Company is considering an Alternative Risk Transfer ("ART") reinsurance solution for capital management purposes. A reinsurer has proposed a solution along the following lines:

- The structure entails reinsurer advancing an initial cash lump-sum to the insurer in exchange for 60% of the future statutory profits on the reinsured block.
- The treaty will cover unit-linked business in force as at 31 March 2019. Any lapsed policies will be covered under the treaty if and when they are reinstated.
- The Company will continue to maintain the underlying unit-linked funds. There will be no change from the investment or asset-liability management standpoint. For the avoidance of doubt, the premiums will not be remitted to the reinsurer and will be invested as per the premium allocation by the policyholders.
- The cash lump-sum from the reinsurer is referred to as 'the Treaty Balance' and accounted for as 'reinsurance commission' in the accounts.
- 60% of the statutory profits from the unit-linked portfolio reinsured will pass through to the reinsurer. This settlement is done quarterly and will be accounted for as 'reinsurance premium' in the accounts. The Treaty Balance will be reduced by the amount of such profits transferred.
- After the Treaty Balance is fully repaid, the reinsurance treaty terminates. Until the Treaty Balance is fully repaid, the Company will also pay the 'Risk Charge', quarterly in arrears. This Risk Charge will also be accounted as 'reinsurance premium' in the accounts.
- The Risk Charge will be calculated as the Treaty Balance times the quarterly Risk Charge rate. The quarterly Risk Charge rate will be one-fourth of (the Mumbai Inter-Bank Offered Rate ("MIBOR") + 3%).
- The treaty has a 'sunset date' of 20 years. It will automatically terminate after 20 years and any Treaty Balance outstanding as of the date of termination will be forfeited.

- ii)** Discuss the implications of the proposed ART solution on the following:

- Solvency ratio
- Statutory reserves
- Embedded Value

Note: You may assume that the Company computes the Embedded Value on an Indian Embedded Value approach.

(8)

iii) Describe how you will justify to the regulator that the proposed ART reinsurance solution is appropriate, commenting on the following aspects:

- Demonstrating ‘significant risk transfer’ (e.g. mortality, lapses)
- Up-front cash lump-sum from the reinsurer
- Investment of the policyholders’ funds
- Benefits of capital efficiency

(15)

The reinsurer has requested the Company to share its expense allocation and apportionment policy.

iv) State the minimum items that are required to be covered in the policy for allocation of direct expenses and apportionment of indirect expenses in accordance with the IRDAI (Expenses of Management of Insurers transacting life insurance business) Regulations, 2016.

(3)

v) Briefly discuss why the reinsurer may be interested to know more about the Company’s expense policy.

(3)

vi) Describe various ways, other than the proposed ART reinsurance solution, that the Company can consider to meet its future capital requirements, commenting on the relative quantum of capital that can be raised, cost of financing and other considerations.

(16)

[50]

Q. 2) The Global Strategy Officer of a multinational insurance company (MNC) is preparing a preliminary ‘go’/’no-go’ assessment based on an evaluation of the core fundamentals of the Indian life insurance market and potential long term benefits to the MNC from investing in the market. The MNC has sought your views.

Globally, the MNC specializes in selling protection business and would like to replicate its international strategy in India with a sole-focus on the following lines of business (LoB):

- Long-term protection covers sold to individuals (including term assurance and whole of life);
- Short-term one year group protection covers sold to corporates or affinity groups;
- Reducing term assurances, linked to either long-term mortgages or shorter term consumer loans sold via lenders and mortgage providers; and
- Immediate life annuities sold to retirees and pensioners.

i) Discuss the current business environment in India in the context of the above product strategy, highlighting the key opportunities and challenges that the Global Strategy Officer should consider when preparing the preliminary ‘go’/’no-go’ assessment for the potential market entry.

(18)

ii) The Global Strategy Officer has asked one of the internal actuaries to prepare a projection of the minimum statutory capital requirements of the insurer, if it is established in India. The internal actuary has computed these using representative model points for expected future new business for a proto-type term assurance (with an option for with and without return of premium at maturity) product. The internal actuary has done so by modifying on of the existing reserving and capital projection models of the MNC in another country. The future reserves and solvency capital requirements are calculated using the following approach:

- Prospective gross premium valuation (GPV) reserves are computed at an individual model point level by discounting future cash-flows;
- Future cash-flows include all outgoes in respect of expenses, commissions and claims (including death, surrender and maturity); and incomes include expected future premium receipts;
- At this stage, the Company does not expect to enter into any reinsurance arrangements and therefore, no allowance is made for reinsurance cash-flows;
- Mortality rates are assumed to be best estimates, increased with a margin for prudence of 10%;
- Lapse/surrender rates are modelled as best estimates varying by policy duration, increased with a margin of 20%;
- Expense loadings are assumed to be the expected long term steady-state ‘per policy’ costs, inflated at a conservative estimate of 6% p.a. (set at the higher end of the central bank’s long term inflation target);
- Net outgo in each projection step (total outgoes less total incomes) are discounted using an estimate of expected future (re)investment rate, set equal to an assumed ultimate risk free forward rate of 6% p.a. for the Indian economy;
- The discounted net present value is zeroised (if negative) at a policy level;
- The solvency capital requirement is computed as 3% of the reserve as computed above plus 0.10% of ‘sum at risk’.

Based on the provisions of IRDAI (Assets, Liability and Solvency Margin) Regulations, 2016 and the applicable actuarial professional standards, comment on any corrections / refinements that might be needed to the above approach to compute the statutory reserving and capital requirements of the Company.

(12)

iii) Whilst undertaking his modelling work for the term assurance product, the internal actuary has made the following two observations:

- that, the typical term assurance premium rates he has observed in India appear to be significantly lower than those observed in other Asian and global markets, even though no such material differences are observed in general population mortality rates; and
- that, despite seemingly “low” premium rates, the value of new business margins from ‘protection’ business in India appears to be higher for protection business (and in particular, for long term assurances) than most other LoBs.

By considering different sources of profit for an individual term assurance product, suggest possible reasons why the above observed outcomes may arise. (10)

- iv)** As an alternative to establishing a new start-up life insurance venture in India, it has been suggested that the MNC may consider acquiring a stake in an existing life insurance company in India. Discuss the relative merits of these two alternative market entry strategies. (10)
- [50]**
